

Program Name: B.Tech (All Branches)	Session.: 2026–27	Semester: I	Section:
Course Name: Physics	Course Code: BAS101	L-T-P-C: 3-1-0-4	Total Lectures: 44
Faculty Name: Dr. S. K. Jain	Designation: Associate Professor	Department: ASH	Contact: 8755169830

No. of Lectures	Topic	Reference	Teaching Method(s)
MODULE I : RELATIVISTIC MECHANICS & INTERFERENCE			
1	1. Frame of reference; inertial and non-inertial frames of reference	T1	C&T
1	2. Galilean transformation and its failure for light waves	T1	C&T
2	3. Michelson–Morley experiment: arrangement, theory and the null result	T1	S/P, C&T
1	4. Postulates of the special theory of relativity	T1	C&T
2	5. Lorentz transformation of space and time (derivation)	T1	C&T
2	6. Length contraction and time dilation (derivation, physical significance)	T1	C&T, QUIZ
1	7. Addition of velocities and variation of mass with velocity	T1	C&T
1	8. Equivalence of mass and energy ($E = mc^2$) and momentum–energy transformation equations	T1	C&T
1	9. Numerical problems and tutorial on relativistic mechanics	T1	QUIZ
1	10. Theory of interference of light; principle of superposition and coherent sources	T2	C&T
1	11. Conditions for obtaining sustained interference	T2	C&T, GD
1	12. Classification of films; interference in a wedge-shaped film (theory)	T2	C&T, CHART
1	13. Fringe width of a wedge-shaped film; applications (testing flatness, thickness measurement)	T2	C&T, DEMO

No. of Lectures	Topic	Reference	Teaching Method(s)
1	14. Numerical problems and tutorial on interference	T2	QUIZ
MODULE II : DIFFRACTION & POLARIZATION			
1	15. Introduction to diffraction; Fresnel and Fraunhofer diffraction	T2	C&T
1	16. Fraunhofer diffraction at a single slit (theory and intensity distribution)	T2	C&T, CHART
1	17. Fraunhofer diffraction at a double slit	T2	C&T
1	18. Diffraction at N slits: the diffraction grating (theory)	T2	C&T, S/P
1	19. Resolving power of a diffraction grating (derivation, Rayleigh criterion)	T2	C&T
1	20. Applications and limitations of a grating; numerical problems	T2	QUIZ
1	21. Phenomena of polarization; double refraction and doubly refracting crystals	T2	C&T, CHART
1	22. Huygens' theory of double refraction (O-ray and E-ray wavefronts)	T2	S/P
1	23. Nicol prism: construction and working as polarizer/analyser	T2	DEMO, C&T
1	24. Quarter-wave plate and half-wave plate	T2	C&T
1	25. Production and analysis of plane, circularly and elliptically polarized light	T2	C&T
1	26. Optical activity; polarimeters (Laurent's half-shade polarimeter)	T2	DEMO
1	27. Applications of polarization; numerical problems and tutorial	T2	QUIZ, GD
MODULE III : LASER, HOLOGRAPHY & FIBER OPTICS			
1	28. Basic principles of laser action: absorption, spontaneous and stimulated emission	T3	C&T
1	29. Population inversion, pumping, the optical resonator, and characteristics of laser light	T3	C&T
1	30. Ruby laser: construction and working (three-level scheme)	T3	S/P, CHART
1	31. He-Ne laser: construction and working (four-level scheme)	T3	S/P, CHART
1	32. Applications of laser	T3	SEM, ET/GL

No. of Lectures	Topic	Reference	Teaching Method(s)
1	33. Basic principle of holography; construction and reconstruction of image on a hologram	T3	S/P
1	34. Applications of holography; tutorial	T3	GD
1	35. Fundamental ideas of optical fiber; types of fibers (step/graded-index, single/multimode)	T3	C&T, CHART
1	36. Acceptance angle, acceptance cone and numerical aperture (derivation)	T3	C&T
1	37. Propagation mechanism in an optical fiber	T3	C&T
1	38. Communication using optical fiber	T3	S/P
1	39. Attenuation and signal loss in optical fiber	T3	C&T
1	40. Dispersion in optical fiber (intermodal and intramodal)	T3	C&T
1	41. Numerical problems, tutorial and overall course revision	T3	QUIZ
44	Total Lectures:		

Teaching Methods: C&T: Chalk & Talk; S/P: Slides/PPT/Videos; SEM: Seminar; DEMO: Demonstration; CHART: Chart/Model; ET/GL: Expert Talk/Guest Lecture; QUIZ: Quiz; GD: Group Discussion; RTCS: Real-Time Case Studies; JAR: Journal Article Review; PD: Poster Design; LAB: Laboratory Demonstration.

Text / Reference Books:

T1. Arthur Beiser, "Concepts of Modern Physics", McGraw-Hill Education.
T2. Ajoy Ghatak, "Optics", Tata McGraw-Hill.
T3. Anuradha De, "Optical Fibre & Laser", New Age International Publishers.
R1. Resnick, Halliday & Walker, "Fundamentals of Physics", Wiley.
R2. R. A. Serway & J. W. Jewett, "Principles of Physics", Cengage (Thomson Asia).